

Elena R. Vazquez

Principal AI Architect

Summary

Microsoft Certified AI Engineer with 12 years of experience designing scalable machine learning infrastructure. Expert in RAG architectures, Azure OpenAI Service, and fine-tuning LLMs for enterprise applications.

Technical Skills

Python, PyTorch, Azure AI Studio, LangChain, Kubernetes, Docker, Terraform, NLP.

Professional Experience

Principal AI Architect | TechGlobal Solutions (2020-Present)

- Designed and deployed a Retrieval-Augmented Generation (RAG) system for a Fortune 500 bank, reducing document search time by 90%.
- Led a team of 10 ML engineers in migrating on-premise inference workloads to Azure Kubernetes Service (AKS).
- Implemented responsible AI guardrails using Azure Content Safety to prevent PII leakage in chatbot responses.

Senior Data Scientist | DataFlow Inc. (2016-2020)

- Developed computer vision models for manufacturing defect detection, achieving 99.5% accuracy.
- Built automated MLOps pipelines using MLflow and Databricks, reducing model deployment time from weeks to days.

Education

M.S. Computer Science, Georgia Tech